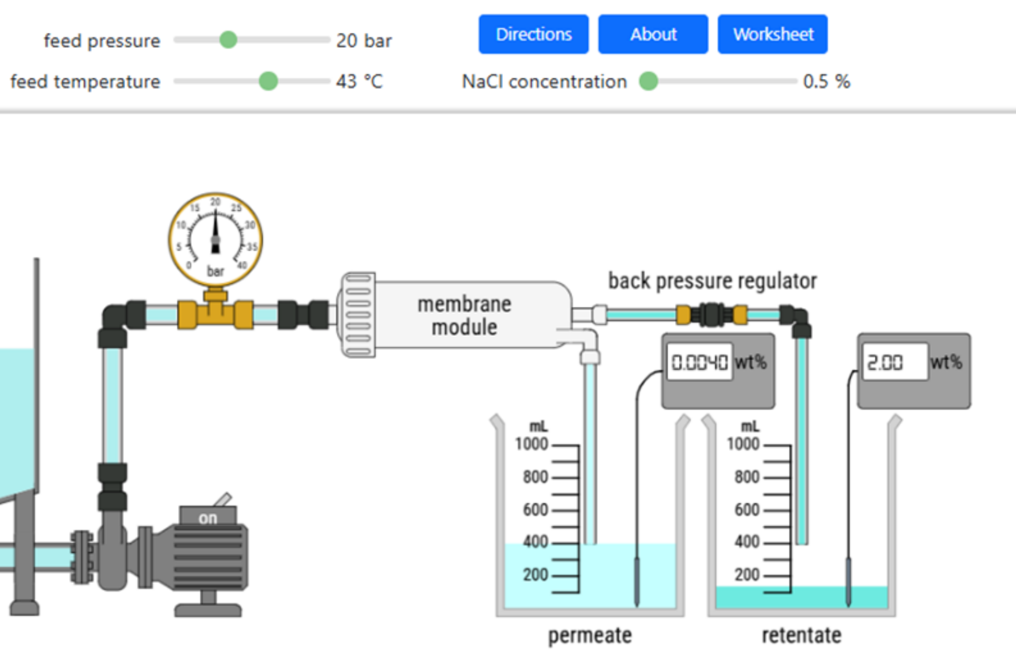
Worksheet: Reverse osmosis

Name(s)____

A reverse osmosis (RO) membrane separates a saltwater solution into a permeate stream (purified water) and a retentate stream (higher salt concentration solution). Reverse osmosis uses high pressure on the feed side (salt water) of a semi-permeable membrane to overcome the osmotic pressure. The experiment examines the effects of feed pressure and salt concentration on permeate flow rate and salt rejection. Reverse osmosis recovers fresh water from sea water on a large scale.

Student Learning objectives

1. Explain the operational principles of reverse osmosis.
2. Calculate permeate flux and salt rejection for a RO membrane.
3. Observe the effects of pressure and feed concentration on permeate flow rate and salt rejection.
4. Apply mass balances to a reverse osmosis system.



Equipment

- A feed tank containing a 0.5-3.5% NaCl solution.
- A high-pressure pump and a pressure gauge that measures the pressure applied to the feed side of the RO unit. The permeate pressure is 1.0 bar.
- A lab-scale RO system with a feed inlet, a semi-permeable membrane (5.0×10^3 cm^2 surface area), and permeate and retentate outlets.

- A back-pressure regulator at the exit of the retentate stream from the module controls the pressure on the feed side.
- Beakers to collect the permeate and retentate streams.
- Conductivity meters measure salt concentrations in the permeate and retentate streams.

Reverse osmosis occurs when the pressure applied exceeds the osmotic pressure difference across the membrane. Water moves from the high-salt feed side to the low-salt permeate side only when the pressure driving force is large enough.

Questions to answer before starting experiment

1. Before collecting data, sketch the expected trend of permeate flux versus pressure for 0.% and 2.5% NaCl. Which curve is higher? Why?

2. Do you expect the percentage of salt rejected to increase, decrease, or remain nearly constant as the feed pressure increases?

3. For a NaCl feed concentration of 3.0%, is there a minimum feed pressure to collect permeate? Why?

Notation

A = membrane area ($5.0\text{x}10^3$ cm^2)

P_f = feed pressure (bar)

P_p = permeate pressure (1.0 bar)

T = temperature (^{o}C or K)

R = ideal gas constant [(bar L)/(mol K)]

i = number of species formed from NaCl (2)

ΔP = pressure drop across membrane (bar)

Π = osmotic pressure (bar)

C = molar salt concentration (mol/L)

J = permeate flux (mol/(cm^2 s))

Q_p = permeate volumetric flow rate (mL/s)

Q_r = retentate volumetric flow rate (mL/s)

c_f= salt concentration in feed (%)

c_p = salt concentration in permeate (%)

c_r = salt concentration in retentate (%)

c_{avg} = average salt concentration in permeate plus retentate (%)

R_s= salt rejection (%)

Experiments

1. Select a feed pressure of 15 bar with the slider.
2. Select a feed temperature of 15°C with the slider.
3. Select a salt feed concentration of 0.5% NaCl with the slider.
4. Click on the switch on top of the pump to turn it on. Allow the system to pressurize and reach steady state.
5. After collecting measurable amounts of liquid in the permeate and retentate beakers, record the volumes and start a timer. You may want to zoom with the scroll wheel to measure the liquid volumes more accurately.
6. After collecting enough of each solution, record their volumes and the elapsed time. Record the data in Table 1. You may want to use a spreadsheet to do your calculations.
7. Record the permeate and retentate concentrations in Table 1.
8. Reset the system and repeat these measurements at 25, 35, and 40 bar feed pressure and record in Table 1.
9. Reset the system, change the salt feed concentration to 2.5 %, and make measurements at 25, 35, and 40 bar and record in Table 1. Why is 25 bar the lowest suggested pressure instead of 15 bar when the feed concentration is 2.5%?

10. Reset the system, set a feed temperature of 50°C, and make measurements for a feed concentration of 0.5% and feed pressures of 15, 25, 35, and 40 bar, and record the results in Table 2.
11. Calculate the permeate and retentate flow rates from all measurements and record these values in Tables 1 and 2.

Table 1 Measurements at 15ºC

Feed pressure (bar)	Feed conc. %	Time (s)	Permeate starting volume (mL)	Permeate final volume (mL)	Retentate starting volume (mL)	Retentate final volume (mL)	Permeate flow rate (mL/s)	Retentate flow rate (mL/s)	Permeate conc. %	Retentate conc. %
15	0.5									
25	0.5									
35	0.5									
40	0.5									
25	2.5									
35	2.5									
40	2.5									

Table 2 Measurements at 50ºC

Feed pressure (bar)	Feed conc. %	Time (s)	Permeate starting volume (mL)	Permeate final volume (mL)	Retentate starting volume (mL)	Retentate final volume (mL)	Permeate flow rate (mL/s)	Retentate flow rate (mL/s)	Permeate conc. %	Retentate conc. %
15	0.5									
25	0.5									
35	0.5									
40	0.5									

Analysis

1. Calculate the osmotic pressure Π of 0.5% and 2.5% NaCl solutions at 15ºC. Compare these values with the feed pressures used in the experiments. At which pressures would you expect no permeate flow? Note that C is the molar salt concentration.
   $\Pi = iCRT$ where i = 2 for NaCl

2. Record the permeate and retentate salt concentrations (%) in Table 3. Calculate the average salt concentration c_{avg} (%) in the two beakers and enter the values in Table 3. How much do these values differ from the feed concentration?

$$c_{avg} = \frac{Q_p c_p + Q_r c_r}{Q_p + Q_r}$$

3. Calculate the salt rejection (R_s) for each experiment and record in Table 3.

$$R_s = \left(1 - \frac{c_p}{c_f}\right) x100$$

4. Calculate the permeate flux (J) for each experiment and record in Table 3.

$J = \frac{Q_p}{A}$ where Q_p is the permeate volumetric flow rate (mL/s) and A is the membrane area (5.0×10^3 cm^2)

Table 3 Analysis

P_f (bar)	T (°C)	c_f (%)	c_p	c_r	c_{avg} (%)	R_s	J (mL/(s cm^2))
15	15	0.5					
25	15	0.5					
35	15	0.5					
40	15	0.5					
25	15	2.5					
35	15	2.5					
40	15	2.5					
15	50	0.5					
25	50	0.5					
35	50	0.5					
40	50	0.5					

5. Plot the permeate flux and salt rejection versus the pressure difference across the membrane (P_f- P_p) for the two salt feed concentrations at 15ºC. Discuss how increasing pressure impacts water flux and salt rejection.

6. Plot the permeate flux versus (pressure difference – osmotic pressure) for the two salt concentrations at 15ºC. What do you observe?

7. Compare the measurements at 15ºC and 50ºC. As the temperature increased, the water viscosity decreased and the osmotic pressure increased. Which effect is more important?

Questions to answer

1. How do higher salt concentrations affect the permeate flux and salt rejection?

2. What safety precautions would you take to conduct this experiment in the laboratory?

3. What other factors should be considered to conduct this process on a large scale?